Highly Efficient Labelling of Extracellular Vesicles for Enhanced Detection on a Microfluidic Platform

*Shi Hu, Rui Hao, Zitong Yu, Huitao Zhang, Hui Yang**
Bionic Sensing and Intelligence Center, Institute of Biomedical and Health Engineering, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China
*Corresponding Author: Hui Yang (hui.yang@siat.ac.cn)

ABSTRACT

Developing precise extracellular vesicle (EV) labelling techniques with minimal disturbance is of great importance to the follow-up EV detection and analysis. Here we introduce a microfluidic platform for enhancing EVs' labelling efficiency. The platform provides an excellent sample throughput and possesses the merits of a higher labelling efficiency with 50% relative improvement and lower background noise compared to conventional co-incubation approach. The characterization results showed that our EV labelling platform could maintain the size and morphology of EVs, holding good potentials in the precise labelling of EVs, which is essential to the subsequent EV detection and analysis.

KEYWORDS: Microfluidic platform, Extracellular vesicles, Labelling

INTRODUCTION

For precise and highly efficient detection of EVs, labelling techniques play an important role. However, the sub-micron size of EVs limits their precise detection and analysis using flow cytometry (FCM). Fluorescent labelling brings the advantages of high sensitivity and selectivity, however, currently available methods such as using probes to conjugate phospholipids or membrane proteins have certain limitations due to steric hindrance, probe aggregation, etc. Microfluidic technology has attracted tremendous attention in bioanalytical applications due to its advantages such as fast response, low sample consumption, and good process control. In this work, we bring a new microfluidic platform labelling technique that can improve EV labelling efficiency and reduce dye aggregation.

EXPERIMENTAL

The microfluidic platform was prepared by plasma bonding a polydimethylsiloxane (PDMS) layer with microchannels and a glass substrate with nanochannels. The PDMS layer consists of 10 channels connected to the inlet and 11 channels connected to outlets, and the glass substrate consists of 1500 parallel nanochannels. After a thorough cleaning, the PDMS layer and the glass substrate were processed by oxygen plasma and bonded, generating an interdigitate layout for subsequent experiments (Figure 1 (A)).

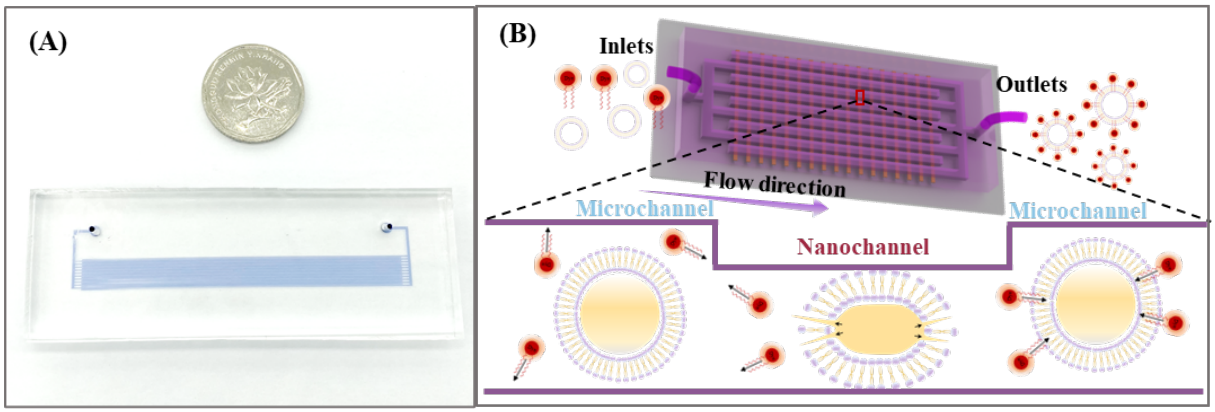

Figure 1: (a) Picture of the EV labelling platform. (b) Schematic illustration of the microfluidic platform for highly efficient EV labelling.

The isolated EVs from human non-small cell lung cancer cells (H1299) were mixed with lipophilic fluorescent dyes (PKH67). Then, the mixed solution was immediately introduced into the microfluidic platform through the inlet at a flow rate of 8 μL/min. Thereafter, the labeled EV samples were collected at the outlet of the platform for the follow-up applications (Figure 1(B)). The EV labelling efficiency was then investigated by FCM, the FCM calibration was conducted with a set of reference silica beads, including 100 nm, 200 nm, 300 nm, and 500 nm beads, respectively.

RESULTS AND DISCUSSION

As shown in Figure 2(A), the EVs display clear membrane structures with round-like and cap-shaped morphologies in the transmission electron microscopy (TEM) image. In addition, the results of nanoparticle tracking analysis (NTA) showed that EVs were mainly classified in the range of 50 nm – 250 nm (Figure 2(B)). According to the statistics of the proportion of EVs in different size ranges, EVs with a size larger than 130 nm account for 54.7% (Figure 3(B)), the nanochannels of our EV labelling platform are approximately 130 nm, which is beneficial to improve the EVs labelling efficiency.

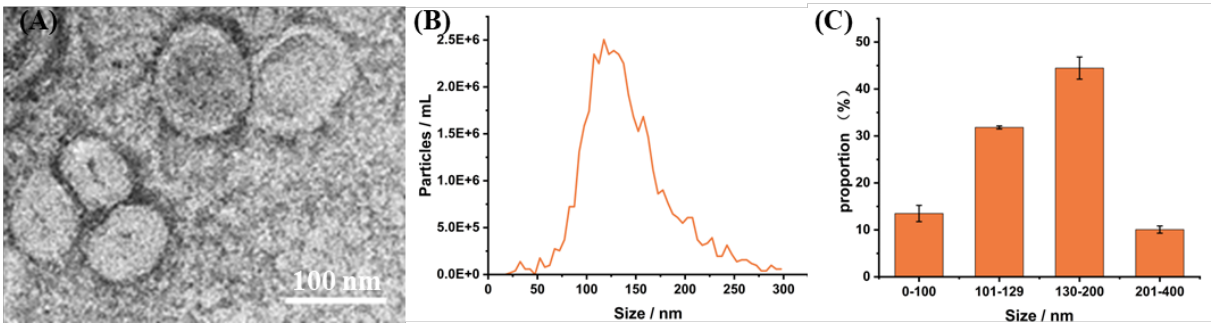

Figure 2: (a) TEM image of the isolated EVs. (b) Size distribution and concentration of the isolated EVs. (c) The histogram on the size of the isolated EVs.

To verify the functionality of the EV labelling platform, a commercial lipophilic dye PKH67 was used in EV labelling efficiency experiments. After the labeled EVs were collected from the outlets, the mixture was analyzed by FCM immediately. In addition, the same volume of incubation group samples was analyzed simultaneously. The experiment group where EVs were mixed with dyes and processed by the microfluidic platform is referred to as Chip, and the experiment group where EVs were passively incubated with fluorescent dyes at room temperature for comparison is referred to as Incubation. At extremely low lipophilic dye concentration, our microfluidic labelling platform can improve EV labelling efficiency and reduce dye aggregation by resetting EVs' phospholipid bilayer arrangement and increasing the probability of dye interacting with the EV membrane. As shown in Figure 3(A)-(B), at low concentrations of PKH67 dyes (0.1%, v/v), the EV labelling efficiency of the chip group is 29.87% and that of the incubation group is 19.97%. By utilizing the microfluidic platform, the EV labelling possesses the merits of a higher labelling efficiency with a relative improvement of 50% compared to conventional co-incubation labelling (Figure 3(C)).

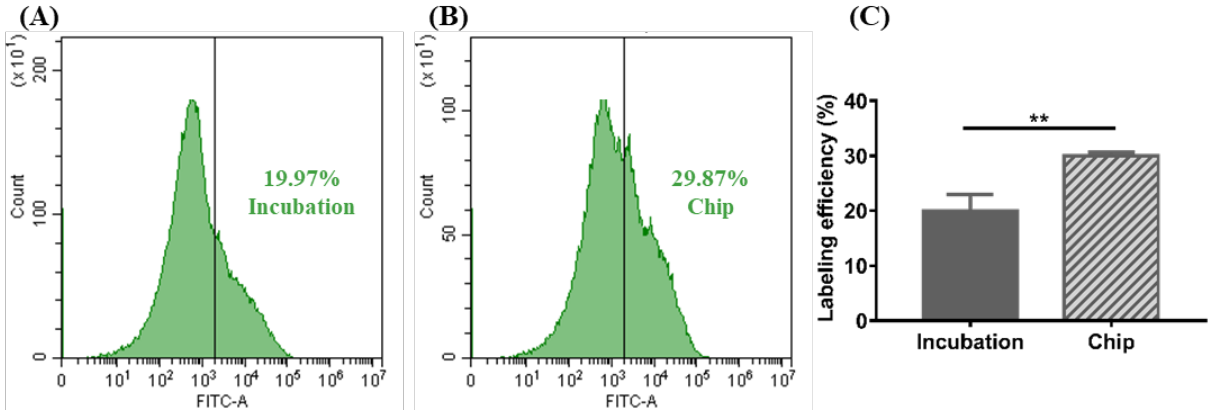

*Figure 3: (A)-(B) FCM detection results of the EV labelling efficiency between the Incubation group and the Chip group. (C) Average labelling efficiency obtained from (A)-(B). (**p < 0.01).*

CONCLUSION

Here, we designed a microfluidic platform that addresses the question of how to reduce false positive signals of dye aggregation when quantifying sub-micron EVs, by resetting the EVs' phospholipid bilayer arrangement and increasing the probability of fluorescent dye interacting with the EV membrane at extremely low dye concentrations. At the optimized flowing rate, our platform has a higher labelling efficiency with a relative improvement of 50% compared to conventional co-incubation labelling. We think that in future, due to the straightforwardness of this approach, the proposed EV labelling technique by utilizing the microfluidic platform will be a robust and versatile tool that can be used to increase the labelling efficiency whilst reducing false positive signal, and eventually benefiting the precise quantification and measurement of EVs.

ACKNOWLEDGEMENTS

We thank the financial support from the National Natural Science Foundation of China (62074155) and Shenzhen Science and Technology Innovation Commission (KCXFZ202002011008124, JSGG20210713091807026).

REFERENCES

[1] Gorgun, Cansu *et al*. "Isolation and flow cytometry characterization of extracellular vesicle subpopulations derived from human mesenchymal stromal cells." Current Protocols in Stem Cell Biology 48.1 (2019): e76.

[2] Yang, Hui *et al*. "Micro-optics for microfluidic analytical applications." Chemical Society Reviews 47.4 (2018): 1391-1458.